



INFOVITS LIMITED

Email: info@infovits.com

Website: <https://infovits.com>

Phone : +265 992 535 274

Address : Area 18A , off interchange behind Puma Filling Station,
Lilongwe Malawi

TECHNICAL & COMMERCIAL PROPOSAL

Mbeu Zanga

Integrated USSD Commerce & Farmer Extension Platform

In Response to RFP Reference: GSM-RFP-2026-001

Submitted to:

Global Seeds Malawi

Submitted by:

InfoVits Ltd

Lilongwe, Malawi

www.infovits.com

Submission Date: 30 January 2026

CONFIDENTIAL

Table of Contents

Table of Contents.....	2
1. Executive Summary.....	6
Why InfoVits Ltd?.....	6
2. Company Profile.....	7
2.1 Company Overview	7
2.2 Company Background	7
2.3 Core Competencies	7
2.4 Technical Team	8
3. Relevant Experience & Case Studies.....	9
3.1 Case Study: MlimiPay Messaging System	9
Project Description.....	9
Scope of Work.....	9
Key Features Delivered	9
Relevance to Mbeu Zanga.....	10
3.2 Case Study: Kwacha Helper USSD & SMS Gateway	10
Project Description.....	10
Scope of Work.....	10
Key Features Delivered	10
Relevance to Mbeu Zanga.....	11
3.3 Case Study: Agritrust Digital Crop Monitoring Platform.....	11
Project Description.....	11
Key Features Delivered	11
Relevance to Mbeu Zanga.....	11
3.4 Case Study: Finance Realm SMS & USSD Integration	11
Project Description.....	12
Key Features Delivered	12
Impact Achieved.....	12
Relevance to Mbeu Zanga.....	12
3.5 Additional Relevant Experience	12
4. Technical Approach.....	14
4.1 Solution Overview	14
4.2 System Architecture.....	14

4.2.1 High-Level Architecture	14
4.2.2 Core Microservices.....	16
4.2.3 Technology Stack	18
4.3 USSD Architecture.....	19
4.3.1 Session Management.....	19
4.3.2 Menu Structure.....	19
4.3.3 Sample User Journey: Seed Purchase via Lay-by	19
4.4 Lay-by Logic & Implementation	21
4.4.1 Lay-by Business Rules.....	21
4.4.2 Lay-by State Machine.....	21
4.4.3 Payment Allocation Logic.....	22
4.5 Mobile Money API Integration Strategy	23
4.5.1 Integration Architecture	23
4.5.2 Airtel Money Integration	23
4.5.3 TNM Mpamba Integration	23
4.5.4 Transaction Lifecycle.....	23
4.5.5 Error Handling & Retry Logic.....	24
4.6 Scalability Strategy	25
4.6.1 Peak Season Considerations	25
4.6.2 Performance Targets.....	25
4.6.3 Load Testing Plan	25
4.7 Security Considerations	27
4.7.1 Security Architecture	27
4.7.2 Role-Based Access Control (RBAC).....	27
4.7.3 Compliance Considerations	28
4.8 Backend Management Portal	29
4.8.1 Portal Overview.....	29
4.8.2 Dashboard & Analytics.....	29
4.8.3 Core Modules.....	29
4.9 Language Support	29
4.9.1 Bilingual Implementation.....	29
4.10 Hosting Recommendation	30
4.10.1 Proposed: Hybrid Architecture	30
5. Project Timeline	31
5.1 Development Phases	31

5.2 Detailed Timeline	31
5.3 Key Milestones	32
6. Commercial Proposal	33
6.1 Implementation & Setup Costs	33
6.1.1 Development & Setup Breakdown	33
6.2 Recurring Fees.....	34
6.2.1 Monthly Recurring Costs.....	34
6.2.2 Annual Recurring Costs	34
6.2.3 Transaction-Based Costs (Pass-Through).....	34
6.3 Annual Cost Summary.....	35
6.4 Payment Schedule.....	36
6.5 Service Level Agreement Coverage.....	36
7. Deliverables.....	37
7.1 Documentation Deliverables	37
7.2 Software Deliverables	37
7.3 Training Deliverables.....	37
8. Service Level Agreement.....	39
8.1 Support Coverage	39
8.2 SLA Inclusions.....	39
8.3 Uptime Guarantee	39
9. Assumptions & Dependencies	40
9.1 Assumptions.....	40
9.2 Dependencies.....	40
10. Conclusion.....	41
Contact Information.....	41
Appendices.....	42
Appendix A: Database Schema Overview	42
Core Entities	42
Appendix B: API Endpoint Summary.....	42
USSD Gateway Endpoints	43
Farmer Management Endpoints	43
Commerce Endpoints.....	43
Payment Endpoints	44
Appendix C: Glossary of Terms	44
Appendix D: References.....	44

1. Executive Summary

InfoVits Ltd Previously called **Infocus Technologies Malawi** is pleased to submit this proposal in response to Global Seeds Malawi's Request for Proposal (RFP Reference: GSM-RFP-2026-001) for the development of the Mbeu Zanga USSD Commerce and Farmer Extension Platform. InfoVits Ltd is the rebranded company name of Infocus Technologies Malawi, in this document, the name InfoVits will be consistently used.

With over nine years of specialized experience in developing USSD platforms, mobile money integrations, and agricultural technology solutions across Malawi, InfoVits Ltd is uniquely positioned to deliver a robust, scalable, and user-centric platform that will transform how smallholder farmers access quality seed and agricultural knowledge.

Why InfoVits Ltd?

- **Proven Track Record:** Successfully deployed multiple USSD platforms processing millions of transactions annually for microfinance institutions, telecommunications companies, and agricultural programs.
- **Local Expertise:** Deep relationships with TNM and Airtel Malawi, with active integrations for USSD gateway and mobile money APIs.
- **Agricultural Domain Knowledge:** Experience building village banking platforms serving rural farming communities across multiple districts.
- **End-to-End Capability:** Full-stack development team capable of delivering from system design through deployment, training, and ongoing support.

Our proposed solution leverages a microservices architecture designed for the unique demands of agricultural commerce, including seasonal traffic spikes during planting periods (October-December), multi-language support (English and Chichewa), and offline-first design principles that ensure accessibility for farmers using basic feature phones in areas with intermittent connectivity.

We are committed to delivering a platform that not only meets but exceeds Global Seeds Malawi's technical requirements while contributing to the broader mission of strengthening food security in Malawi through improved farmer access to quality seed and agronomic knowledge.

2. Company Profile

2.1 Company Overview

Legal Name	InfoVits Ltd (formerly InFocus Technologies)
Registration	Registered under the Companies Act of Malawi
Year Established	2019
Headquarters	Lilongwe, Malawi
Website	www.infovits.com / www.infocustech-mw.com
Primary Contact	Ngina Kamuloni
Email	info@infovits.com / info@infocustech-mw.com / ngina@infovits.com
Phone	+265 994 89 41 48

2.2 Company Background

InfoVits Ltd is a leading Malawian software development company specializing in financial technology, telecommunications solutions, and enterprise management systems. Since our establishment in 2019, we have grown to become a trusted technology partner for financial institutions, telecommunications operators, government agencies, and international development organizations operating in Malawi.

Our core competencies span the development of USSD platforms, mobile money integrations, core banking systems, microfinance management solutions, and large-scale data processing systems. We pride ourselves on delivering solutions that are specifically designed for the Malawian context, taking into account local infrastructure constraints, regulatory requirements, and user needs.

2.3 Core Competencies

Domain	Expertise	Years
USSD Development	Gateway integration, session management, menu design	7+
Mobile Money	Airtel Money, TNM Mpamba API integration	8+

Domain	Expertise	Years
Fintech	Core banking, microfinance, payment processing	9+
Agricultural Tech	Farmer registration, village banking, extension services	5+
Telecommunications	SMS gateways, bulk messaging, SMPP integration	9+
Enterprise Systems	ERP, case management, workflow automation	7+

2.4 Technical Team

InfoVits Ltd maintains a skilled technical team with expertise across the full technology stack required for this project. Our team includes senior software developers with extensive experience in Node.js, C#,PHP (Laravel, CodeIgniter), React, Blazor and database technologies including PostgreSQL , **MSSQL** and MySQL.

Role	Expertise	Relevant Experience
Lead Developer	C#, Node.js, PHP, System Architecture	9 years in fintech & USSD development
Backend Developer	API Development, Database Design	6+ years in payment systems
Frontend Developer	React, Vue.js, Dashboard Development	5+ years in web applications
DevOps Engineer	Linux, Docker, Server Administration	7+ years infrastructure management
QA Engineer	Testing, Security Auditing	5+ years in financial technology

3. Relevant Experience & Case Studies

InfoVits Ltd has successfully delivered numerous projects that directly relate to the requirements of the Mbeu Zanga platform. Our team brings deep expertise in agricultural technology, USSD development, and mobile money integration—including foundational work on village banking platforms where our directors served as lead developers. Below we present case studies demonstrating our proven capabilities.

3.1 Case Study: MlimiPay Messaging System

Client	CASA in collaboration with TEREN Agro
Project Type	Agricultural Advisory & Farmer Engagement Platform
Status	Live and Operational
Technology	SMS Gateway, Mobile Money Integration, Farmer Database

Project Description

Design, configuration, and deployment of a robust messaging system to enhance smallholder farmer engagement through improved digital advisory services and communication. The platform serves thousands of farmers across Malawi with timely agricultural information.

Scope of Work

- System analysis and requirements gathering
- Messaging platform development and deployment
- Integration with MlimiPay's existing infrastructure
- Testing, quality assurance, and go-live support
- User training and comprehensive documentation

Key Features Delivered

- Automated messaging for farmer engagement and advisory services
- Integration with mobile money platforms for transactional notifications
- Real-time delivery of farming tips and seasonal updates
- Accessible platform design for farmers of varying literacy levels
- Data security and compliance with privacy regulations

Relevance to Mbeu Zanga

MlimiPay demonstrates our direct experience building agricultural communication platforms for smallholder farmers in Malawi. The automated messaging, farmer database management, and accessibility considerations translate directly to Mbeu Zanga's extension services and farmer engagement requirements.

3.2 Case Study: Kwacha Helper USSD & SMS Gateway

Client	Kwacha Helper
Project Type	USSD Gateway & Automated Loan Management Integration
Status	Live - Processing daily loan transactions
Technology	USSD Gateway, SMS Gateway, Airtel Money API

Project Description

USSD and SMS gateway integration with Kwacha Helper's automated loan management system, enabling digital loan disbursement, collection, and real-time customer communication through mobile channels.

Scope of Work

- USSD gateway development , integration ,configuration
- SMS gateway integration for notifications
- Airtel Money API integration for automated payments
- API development for loan management system connectivity
- Comprehensive testing and production deployment
- Infrastructure setup (Network, VPN, Internet, servers config and more)

Key Features Delivered

- USSD-based loan application and status enquiry
- Automated SMS notifications for loan approvals, disbursements, and repayment reminders
- Airtel Money integration for automated loan disbursement and collection
- Real-time session management and transaction logging
- Secure API communication between gateway and loan management system

Relevance to Mbeu Zanga

Kwacha Helper showcases our expertise in building complete USSD self-service platforms with mobile money integration. The payment automation, session management, and notification systems directly mirror the requirements for Mbeu Zanga's lay-by payment and order tracking functionality.

3.3 Case Study: Agritrust Digital Crop Monitoring Platform

Client	Agritrust
Project Type	Digital Crop Monitoring & SMS Advisory Platform
Status	Live and Operational
Technology	Web Platform, SMS Gateway, Farmer Database

Project Description

Development of a digital crop monitoring platform with automated SMS notifications to support farmers throughout their farming cycles, delivering timely advisory information aligned with crop growth stages.

Key Features Delivered

- Automated crop monitoring and tracking across farming stages
- SMS alerts triggered by farming stages (planting, fertiliser application, harvesting)
- Farmer database management with registration and profiling
- Reporting and analytics dashboard for programme monitoring

Relevance to Mbeu Zanga

Agritrust directly demonstrates our capability to build agricultural extension platforms with stage-based notifications—exactly what Mbeu Zanga requires for delivering crop-specific agronomic tips, weather alerts, and seasonal guidance to farmers.

3.4 Case Study: Finance Realm SMS & USSD Integration

Product	Finance Realm Loan Management System (InfoVits Ltd)
Clients Served	Multiple Microfinance Institutions across Malawi
Status	Live - Multi-tenant production deployment

Technology

USSD (TNM & Airtel), SMS Gateway, Mobile Money APIs

Project Description

Integration of SMS and USSD capabilities into Finance Realm, a comprehensive microfinance management system developed by InfoVits Ltd, now serving multiple organisations across Malawi with automated financial services delivery.

Key Features Delivered

- USSD gateway integration with both TNM and Airtel networks
- USSD self-service for loan applications, balance enquiries, and repayment schedules
- Automated SMS notifications for complete loan lifecycle events
- Multi-tenant support serving various microfinance institutions
- Integration with mobile money platforms for disbursements and collections
- Comprehensive audit trails and transaction logging

Impact Achieved

- Extended financial services reach to underserved populations
- Reduced operational costs through automation
- Improved client engagement and loan recovery rates

Relevance to Mbeu Zanga

Finance Realm demonstrates our ability to build and operate multi-tenant platforms with dual-network USSD integration (TNM and Airtel), mobile money payment processing, and automated notifications—all core requirements for the Mbeu Zanga platform.

3.5 Additional Relevant Experience

Beyond the case studies above, our team brings foundational expertise from prior work on mobile money and financial inclusion platforms. The directors of InfoVits Ltd served as the original developers of KakuPay, a mobile money wallet platform that processes sending and receiving money, bill payments, and airtime purchases. KakuPay also features specialised modules including a village banking module for savings groups and a social cash transfer module that has disbursed billions of Kwacha to beneficiaries in rural communities. This deep experience in building comprehensive financial platforms for diverse user segments directly informs our approach to the Mbeu Zanga platform.

Project	Client Type	Key Features	Relevance
Maso Athu Election Platform	Civic Organization	Real-time USSD data collection, analytics dashboard	Large-scale USSD deployment
ECMS	UNDP/Government	Case management, workflow automation, multi-tenant	Enterprise architecture
KakuPay Mobile Wallet	Fintech	Mobile money, village banking module, social cash transfer	Financial services at scale

4. Technical Approach

4.1 Solution Overview

Our proposed solution for the Mbeu Zanga platform is built on a microservices architecture designed for scalability, reliability, and maintainability. The system separates concerns across distinct service modules, allowing independent scaling of high-demand components during peak agricultural seasons and enabling continuous deployment of improvements without system-wide downtime.

4.2 System Architecture

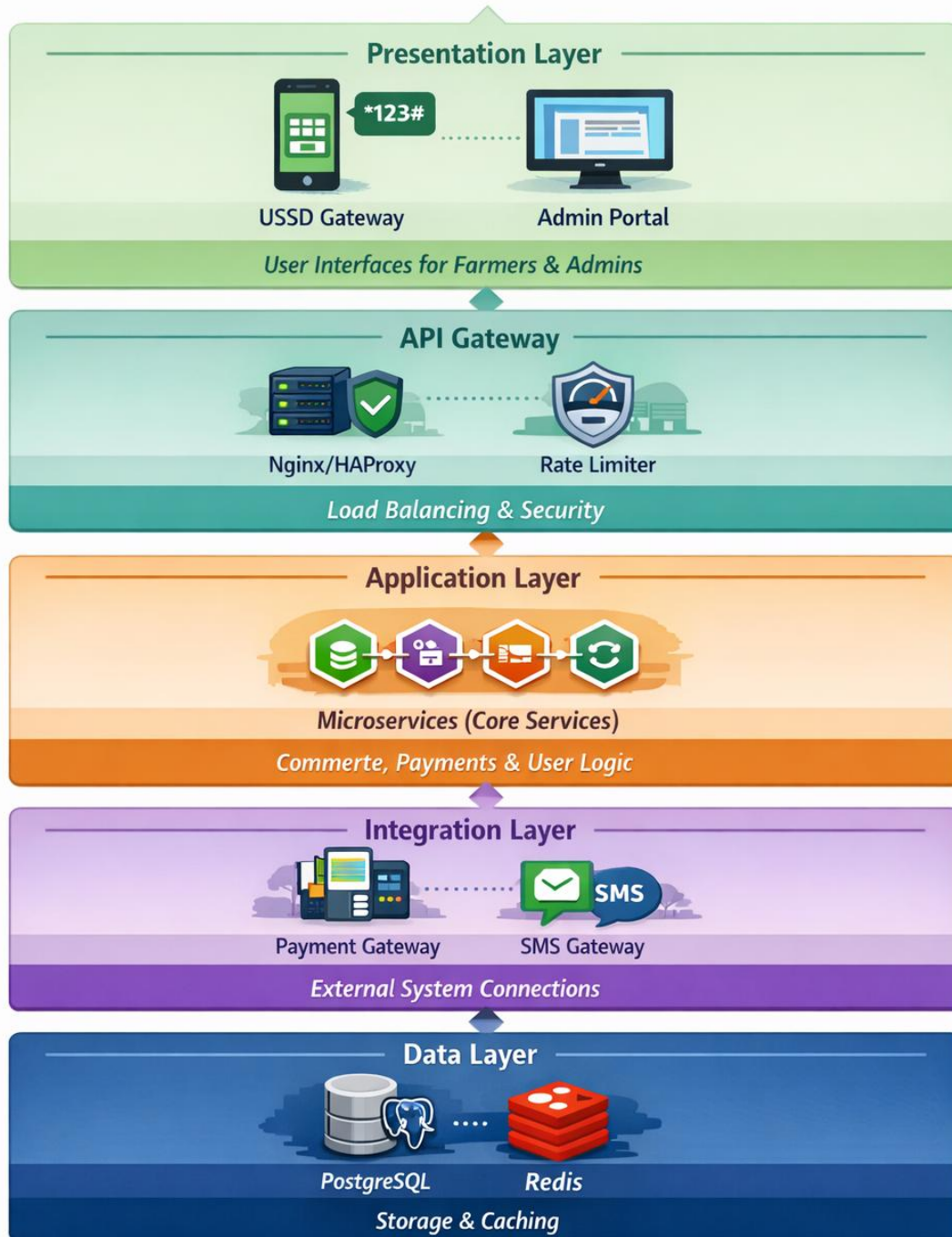
4.2.1 High-Level Architecture

The platform consists of five logical layers, each designed for specific responsibilities. This separation ensures that the system can scale horizontally to handle seasonal traffic spikes while maintaining consistent performance and reliability.

Layer	Components	Purpose
Presentation Layer	USSD Gateway, Admin Portal	User interfaces for farmers (USSD) and administrators (Web)
API Gateway	Nginx/HAProxy, Rate Limiter	Load balancing, request routing, security enforcement
Application Layer	Microservices (6 core services)	Business logic for commerce, payments, extensions, users
Integration Layer	Payment Gateway, SMS Gateway	External system connections (Airtel, TNM, SMS providers)
Data Layer	PostgreSQL, Redis	Persistent storage and high-speed caching

4.2.1 High-Level Architecture

The platform consists of five logical layers, each designed for specific responsibilities. This separation ensures that the system can scale horizontally to handle seasonal traffic spikes while maintaining consistent performance and reliability.



4.2.2 Core Microservices

The application layer is divided into six specialized microservices, each handling a distinct domain of functionality. This modular approach allows for independent development, testing, and deployment of each component.

Service	Responsibility	Key Functions
Session Service	USSD session management	State machine, menu navigation, timeout handling, language preferences
User Service	Farmer registration & profiles	Registration, authentication, profile management, district mapping
Commerce Service	Product catalog & orders	Inventory lookup, pricing, order creation, collection point assignment
Payment Service	Transaction processing	Mobile money integration, lay-by management, reconciliation
Extension Service	Agricultural content delivery	Content library, push notifications, weather alerts, scheduling
Notification Service	SMS & messaging	Transactional SMS, bulk campaigns, delivery tracking

Core Microservices

The application layer is divided into six specialized microservices, each handling a distinct domain of functionality. This modular approach allows for independent development, testing, and deployment of each component.

Service	Responsibility	Key Functions
	Session Service USSD session management	• State machine, menu navigation • Timeout handling • Language preferences
	User Service Farmer registration & profiles	• Registration, authentication • Profile management • District mapping
	Commerce Service Product catalog & orders	• Inventory lookup, pricing • Order creation • Collection point assignment
	Payment Service Transaction processing	• Mobile money integration • Lay-by management • Reconciliation
	Extension Service Agricultural content delivery	• Content library • Push notifications • Weather alerts • Scheduling
	Notification Service SMS & messaging	• Transactional SMS • Bulk campaigns • Delivery tracking

Independent Development, Testing & Deployment

4.2.3 Technology Stack

Component	Technology	Justification
Runtime	Node.js 20 LTS	High I/O throughput ideal for USSD; extensive async support
Framework	Express.js / Fastify	Lightweight, proven for high-volume API services
Primary Database	PostgreSQL 16 / MySQL >8	ACID compliance for financial data; advanced JSON support
Cache Layer	Redis Cluster	Sub-millisecond session retrieval; distributed locking
Message Queue	RabbitMQ	Reliable async processing for payments and notifications
Admin Portal	React.js	Modern SPA framework; component reusability
API Documentation	OpenAPI 3.0 / Swagger	Industry-standard API documentation
Containerization	Docker	Consistent deployment across environments
Orchestration	Docker Swarm / Kubernetes	Auto-scaling, service discovery, rolling updates

4.3 USSD Architecture

4.3.1 Session Management

USSD interactions are inherently stateful and time-constrained. Our architecture uses Redis for ultra-fast session storage with automatic expiration handling. Each session maintains a state machine that tracks the user's position in the menu hierarchy, input history, and transaction context.

Key session management features include automatic session recovery when users dial back within the timeout window, graceful handling of network interruptions through transaction savepoints, and multi-device session blocking to prevent concurrent access from different phones.

4.3.2 Menu Structure

The USSD menu is designed following best practices for accessibility and usability. Menu depth is limited to five levels maximum, with the most common operations (check balance, make payment) accessible within three selections. All menus support navigation shortcuts (0 for back, 00 for main menu).

Main Menu (*XXXX#):

1. Register / Update Profile (Lembani)
2. Buy Seed (Gula Mbeu)
3. My Lay-by (Malipiro Anga)
4. My Orders (Ma Oda Anga)
5. Farming Tips (Uphungu wa Ulimi)
6. Help / Change Language

4.3.3 Sample User Journey: Seed Purchase via Lay-by

The following illustrates a typical farmer journey for purchasing seed through the lay-by option:

Step	Screen Content	Input
1	Welcome to Global Seeds. Select: 1.Register 2.Buy Seed 3.Lay-by 4.Orders 5.Tips	2
2	Select Category: 1.Maize 2.Legumes 3.Horticulture 0.Back	1
3	Maize: 1.SC403 (10kg)-MK25,000 2.DK8053 (10kg)-MK28,000 3.MH26 (5kg)-MK15,000	1
4	SC403 10kg @ MK25,000. Quantity (bags)?	2
5	Total: MK50,000. Pay: 1.Full Amount 2.Lay-by (30% deposit)	2

Step	Screen Content	Input
6	Lay-by Plan: Deposit MK15,000 + 3 monthly payments of MK11,667. Confirm? 1.Yes 2.No	1
7	Pay via: 1.Airtel Money 2.TNM Mpamba	1
8	Enter Airtel Money PIN to pay MK15,000	[PIN]
9	Deposit received! Order #GSM2026001. Balance MK35,000. SMS sent.	—

4.4 Lay-by Logic & Implementation

4.4.1 Lay-by Business Rules

The lay-by system enables farmers to reserve seed through installment payments, improving affordability and pre-season planning. The following configurable rules govern lay-by transactions:

Parameter	Default Value	Configurable
Minimum Deposit	30% of order value	Yes (20-50%)
Maximum Lay-by Duration	6 months	Yes (3-12 months)
Payment Frequency Options	Weekly, Bi-weekly, Monthly	Yes
Grace Period	7 days after due date	Yes (0-14 days)
Late Payment Notification	1 day, 3 days, 7 days after due	Yes
Default Threshold	30 days overdue	Yes (14-60 days)
Forfeiture Policy	Configurable (refund/forfeit)	Yes
Early Completion Discount	Optional 5% discount	Yes (0-10%)

4.4.2 Lay-by State Machine

Each lay-by order progresses through a defined lifecycle with clear transitions. The system enforces state transitions and triggers appropriate notifications at each stage.

State	Description	Triggers
INITIATED	Order created, awaiting deposit	Farmer confirms lay-by selection
DEPOSIT_PAID	Initial deposit received	Successful deposit payment callback
IN_PROGRESS	Active payment schedule	At least one installment after deposit
COMPLETED	Full payment received	Final installment clears balance to zero
READY_FOR_COLLECTION	Stock reserved, awaiting pickup	Admin marks as ready OR automatic
COLLECTED	Farmer received goods	Admin confirms collection

State	Description	Triggers
OVERDUE	Payment missed beyond grace period	Cron job detects missed payment
DEFAULTED	Extended non-payment	30+ days overdue (configurable)
CANCELLED	Order cancelled	Admin action OR farmer request before collection

4.4.3 Payment Allocation Logic

When a payment is received, the system applies funds using a priority-based allocation algorithm. This ensures transparent and predictable application of farmer payments.

7. Apply to oldest overdue installment first (FIFO principle)
8. If amount exceeds installment due, apply remainder to next scheduled payment
9. Update remaining balance and recalculate future installments if overpaid
10. Trigger SMS confirmation with updated balance
11. If balance reaches zero, transition order to COMPLETED state

4.5 Mobile Money API Integration Strategy

4.5.1 Integration Architecture

Mobile money integration is critical to the platform's success. We implement a unified payment gateway abstraction layer that provides a consistent interface for both Airtel Money and TNM Mpamba, simplifying business logic and enabling easy addition of future payment providers.

The abstraction layer handles provider-specific authentication, request formatting, callback parsing, and error handling. This approach has been proven in our production systems handling thousands of daily transactions.

4.5.2 Airtel Money Integration

Capability	API Endpoint	Use Case
Collection Request	POST /merchant/v2/payments/	Initiate USSD payment prompt to farmer
Transaction Status	GET /standard/v2/payments/{id}	Query payment confirmation
Callback Handler	POST /webhooks/airtel	Receive real-time payment notifications
Refund	POST /merchant/v2/refund/	Process cancellation refunds

4.5.3 TNM Mpamba Integration

Capability	API Endpoint	Use Case
Payment Request	POST /api/v1/payment/request	Push payment prompt to farmer
Status Check	GET /api/v1/payment/status/{ref}	Verify transaction completion
Callback Handler	POST /webhooks/tnm	Receive payment confirmations
Balance Inquiry	GET /api/v1/merchant/balance	Monitor float balance

4.5.4 Transaction Lifecycle

To ensure data integrity and prevent duplicate processing, all payment transactions follow an idempotent processing pattern with the following safeguards:

- **Unique Reference Generation:** Each transaction receives a unique reference combining timestamp, farmer ID, and random component
- **Idempotency Keys:** Callbacks are processed only once; duplicate callbacks are acknowledged but not reprocessed
- **Database Transactions:** Payment updates and balance adjustments occur within atomic database transactions
- **Reconciliation Jobs:** Automated daily reconciliation compares local records with provider statements

4.5.5 Error Handling & Retry Logic

Scenario	Handling Strategy	User Experience
Timeout on payment request	Queue for retry (3 attempts, exponential backoff)	SMS notification of pending status
Insufficient funds	Return clear error, suggest partial payment	USSD message with balance advice
Provider downtime	Failover to alternate provider if available	Inform user of temporary issue
Callback not received	Scheduled status polling (5min intervals)	Automatic confirmation when resolved
Duplicate callback	Acknowledge, skip processing	No duplicate SMS or charges

4.6 Scalability Strategy

4.6.1 Peak Season Considerations

Agricultural commerce follows predictable seasonal patterns, with the October-December planting period generating 10-20x normal transaction volumes. Our architecture is designed specifically to handle these traffic spikes without degradation in performance or reliability.

Strategy	Implementation	Benefit
Horizontal Auto-scaling	Kubernetes HPA based on CPU/request metrics	Automatic capacity adjustment during peaks
Database Read Replicas	PostgreSQL streaming replication	Distribute read load, maintain write performance
Connection Pooling	PgBouncer with transaction-mode pooling	Efficient database connection management
Distributed Caching	Redis Cluster (3+ nodes)	Session access <1ms regardless of scale
Async Processing	RabbitMQ for payments & notifications	Non-blocking request handling
Static Content CDN	CloudFlare or local CDN	Reduce origin server load

4.6.2 Performance Targets

Metric	Target	Measurement Method
USSD Response Time	<500ms (95th percentile)	Application performance monitoring
Concurrent Sessions	5,000+ simultaneous	Load testing with realistic scenarios
Payment Processing	100 TPS sustained	Provider API limits permitting
Database Query Time	<50ms (average)	Query profiling and optimization
System Availability	99.9% uptime	Monitoring and alerting

4.6.3 Load Testing Plan

Prior to production deployment, we will conduct comprehensive load testing to validate system performance under realistic conditions. Testing scenarios will include sustained load (normal operations), spike testing (sudden traffic increases), and stress testing (beyond expected maximum).

4.7 Security Considerations

4.7.1 Security Architecture

Security is paramount for a platform handling financial transactions and personal farmer data. Our multi-layered security approach addresses threats at each system layer.

Layer	Security Measures	Implementation
Network	TLS 1.3, firewall rules, DDoS protection	All external communications encrypted; IP whitelisting for admin
Application	Input validation, rate limiting, OWASP compliance	Strict validation; per-MSISDN rate limits; security scanning
Data	Encryption at rest, column-level encryption, secure backups	AES-256 for sensitive fields; encrypted backup storage
Authentication	MSISDN verification, admin 2FA, session tokens	USSD inherent auth; time-based OTP for admin portal
Audit	Immutable transaction logs, access logging	Append-only audit tables; tamper-evident logging
VPN	Secure tunnel between Our servers and MNOs	All traffic is encrypted and not passing through direct HTTPs

4.7.2 Role-Based Access Control (RBAC)

The administrative portal implements granular role-based access control to ensure staff members only access functions appropriate to their responsibilities.

Role	Permissions	Typical Users
Super Admin	Full system access, user management, configuration	IT Manager, System Owner
Finance	Payment reconciliation, financial reports, settlements	Accountants, Finance Staff
Operations	Inventory management, order processing, farmer data	Warehouse Staff, Customer Service
Extension	Content management, SMS campaigns, analytics	Agricultural Officers
Support	Read-only farmer queries, order status lookup	Call Center Staff

Role	Permissions	Typical Users
Auditor	Read-only access to all logs and reports	Internal/External Auditors

4.7.3 Compliance Considerations

- **MACRA Compliance:** USSD service registration and compliance with MACRA guidelines and data protection policies
- **RBM Guidelines:** Adherence to Reserve Bank of Malawi requirements for mobile money aggregation
- **Data Protection:** Alignment with emerging Malawi data protection framework and GDPR-inspired best practices

4.8 Backend Management Portal

4.8.1 Portal Overview

The web-based administration portal provides Global Seeds Malawi staff with comprehensive tools for managing all aspects of the platform. Built using React.js, the portal offers a responsive, intuitive interface accessible from any modern web browser.

4.8.2 Dashboard & Analytics

The main dashboard provides real-time visibility into platform performance with key metrics including daily transaction volumes, revenue by product category, active lay-by summary with at-risk accounts flagged, farmer registration trends by district, and inventory status with low-stock alerts.

4.8.3 Core Modules

Module	Key Features
Inventory Management	Product catalog CRUD, stock levels by collection point, pricing with effective dates, low-stock alerts
Order Management	Order listing with filters, lay-by monitoring, collection assignment, delivery confirmation
Payment Reconciliation	Transaction listing, provider reference matching, automated status, manual tools, settlement reports
Extension Content	Content CMS with Chichewa support, campaign builder (district/crop targeting), scheduling, delivery reports
User Management	Staff accounts, role assignment, activity logs, password policies
Reporting	Pre-built reports, custom report builder, export to Excel/PDF, scheduled reports

4.9 Language Support

4.9.1 Bilingual Implementation

All farmer-facing content is available in both English and Chichewa. Language selection is captured during registration and stored in the farmer profile, with the option to change language at any time through the USSD menu.

The system uses an internationalization (i18n) framework that separates content from code, enabling easy content updates and potential addition of future languages. All SMS notifications, USSD menus, and extension content respect the farmer's language preference.

4.10 Hosting Recommendation

4.10.1 Proposed: Hybrid Architecture

We recommend a hybrid hosting approach that balances performance, reliability, and cost-effectiveness. This architecture places latency-sensitive components locally while leveraging cloud infrastructure for scalability and disaster recovery.

Component	Location	Rationale
USSD Gateway Connector	Local VPS (Malawi)	Minimal latency to TNM/Airtel APIs; regulatory compliance
Application Servers	Cloud (AWS/Azure/VPS)	Auto-scaling capability; managed infrastructure
Primary Database	Cloud (Cape Town Region)	Managed PostgreSQL with automated backups
Redis Cache	Cloud (Cape Town Region)	Managed Redis cluster for session storage
Backup/DR	Secondary Cloud Region	Geographic redundancy for business continuity

5. Project Timeline

5.1 Development Phases

The project will be executed in five phases over approximately 12 weeks (3 months), with clear milestones and deliverables at each stage. This timeline assumes prompt contract finalization and timely provision of required information and approvals from Global Seeds Malawi.

Phase	Duration	Key Activities	Deliverables
1. Inception & Planning	2 weeks	Requirements validation, system analysis, SRS documentation	SRS Document, Technical Design, Project Plan
2. Core Development	4 weeks	USSD engine, commerce module, lay-by system, admin portal	Working USSD platform, Admin portal, Core APIs
3. Integration	3 weeks	Mobile money APIs, SMPP setup, shortcode integration	Integrated payment processing, SMS gateway
4. Testing & QA	2 weeks	Integration testing, UAT, load testing, security audit	Test reports, Performance benchmarks
5. Deployment & Training	1 week	Production deployment, staff training, go-live support	Live system, Training materials, Manuals

5.2 Detailed Timeline

Week	Phase	Milestone
Week 1	Inception	Project kickoff, requirements workshop, system analysis
Week 2	Inception	SRS documentation and sign-off
Week 3-4	Core Dev	USSD framework, session management, farmer registration
Week 5-6	Core Dev	Commerce module, lay-by system, admin portal development
Week 7-8	Integration	Airtel Money & TNM Mpamba API integration
Week 9	Integration	SMPP protocol setup, shortcode integration, SMS gateway

Week	Phase	Milestone
Week 10-11	Testing	UAT with GSM staff, bug fixes, load testing
Week 12	Deployment	Production deployment, staff training, go-live

5.3 Key Milestones

Milestone	Target Date	Success Criteria
Contract Signature	16 February 2026	Signed agreement, project initiated
SRS Approval	28 February 2026	Requirements document signed off by GSM
USSD Prototype Demo	14 March 2026	End-to-end USSD flow demonstrated
Payment Integration Complete	4 April 2026	Successful test transactions on both providers
UAT Sign-off	25 April 2026	GSM approval of tested system
Go-Live	9 May 2026	Production system launched

6. Commercial Proposal

6.1 Implementation & Setup Costs

The following budget covers the complete development, integration, and deployment of the Mbeu Zanga platform. All prices are quoted in Malawi Kwacha (MWK). This quotation is valid for implementation within Lilongwe; outside Lilongwe, allowances and travel costs will apply.

6.1.1 Development & Setup Breakdown

Item	Description	Amount (MWK)
System Analysis	Requirements gathering, workflow analysis, technical assessment	1,642,500.00
SRS Documentation	System Requirements Specification, technical design documents	1,857,500.00
Software Development	Core USSD engine, admin portal, backend services, lay-by module	18,440,000.00
Payment Integration	Airtel Money & TNM Mpamba API integration and testing	1,250,000.00
SMPP Protocol Setup	SMS gateway configuration, bulk messaging engine setup	550,000.00
Shortcode Integration & Setup	USSD shortcode registration and telco integration	510,000.00
Documentation	Technical documentation, user manuals, operations guide	750,000.00
Training	Staff training on system administration and operations	3 500,000.00
Project Management	Coordination, progress reporting, quality assurance	1,000,000.00
Subtotal		29,500,000.00
VAT (17.5%)		5,162,500.00
GRAND TOTAL		34,662,500.00.

Note: This quotation is based on implementation within Lilongwe. For deployments or training sessions outside Lilongwe, transport, accommodation, and daily subsistence allowances will be quoted separately based on location and duration.

6.2 Recurring Fees

The following recurring costs apply for ongoing platform operation, maintenance, and support services.

6.2.1 Monthly Recurring Costs

Item	Description	Frequency	Amount (MWK)
Shortcode Rental	USSD shortcode monthly rental fee (telco charge)	Monthly	280,000.00
System Maintenance	Technical support, bug fixes, security patches, monitoring	Monthly	612,000.00
Monthly Subtotal			892,000.00

6.2.2 Annual Recurring Costs

Item	Description	Frequency	Amount (MWK)
System Hosting	Cloud infrastructure, database, caching, backups	Annual	2,500,000.00

6.2.3 Transaction-Based Costs (Pass-Through)

Item	Description	Rate (MWK)
USSD Session Fee	Per USSD session charge (telco rate)	17.00 per session
SMS Messaging Cost	Per SMS notification/campaign message	17.00 per SMS

Note: Transaction-based costs (USSD sessions and SMS) are pass-through charges at telco rates and will be invoiced based on actual monthly usage. Mobile money transaction fees charged by Airtel Money and TNM Mpamba are separate and borne by the transacting party as per provider terms.

6.3 Annual Cost Summary

The following provides an indicative annual operating cost summary based on estimated usage:

Cost Category	Annual Amount (MWK)
Shortcode Rental (12 months)	3,360,000.00
System Maintenance (12 months)	7,344,000.00
System Hosting (Annual)	2,500,000.00
USSD & SMS Costs	Variable (MWK 17/session & MWK 17/SMS)
TOTAL FIXED ANNUAL COSTS	13,204,000.00

Note: The above excludes variable transaction costs (USSD sessions at MWK 17.00 each and SMS at MWK 17.00 each) which will be invoiced based on actual monthly usage.

6.4 Payment Schedule

We propose the following milestone-based payment schedule for the implementation costs:

Milestone	Percentage	Amount (MWK)	Trigger
Contract Signature	30%	9,341,250.00	Upon signing of contract
SRS Approval	20%	6,227,500.00	Sign-off of System Requirements Specification
Development Complete	30%	9,341,250.00	USSD platform and admin portal demonstrated
UAT Sign-off	10%	6,113,750.00	User acceptance testing approval
Go-Live	10%	3 638 750.00	Successful production deployment
Total (incl. VAT)	100%	34,662,500.00	

6.5 Service Level Agreement Coverage

The monthly System Maintenance Fee of MWK 612,000 includes the following support services:

- Technical support via email, phone, and ticketing system (business hours)
- Bug fixes and patches for reported defects
- Security updates and vulnerability patches
- 24/7 automated system monitoring with proactive alerting
- Database backup verification and recovery testing
- Up to 20 hours of minor enhancements/changes per month
- Monthly system health reports
- Quarterly review meetings

7. Deliverables

7.1 Documentation Deliverables

Document	Description	Delivery Phase
System Requirements Specification	Detailed functional and non-functional requirements	Phase 1
Technical Design Document	Architecture, database schema, API specifications	Phase 1
API Documentation	OpenAPI/Swagger documentation for all endpoints	Phase 2-4
Database Documentation	ERD, data dictionary, backup procedures	Phase 2
Security Documentation	Security architecture, access control matrix	Phase 5
Operations Manual	Deployment procedures, monitoring, troubleshooting	Phase 6
User Manual (Admin Portal)	Guide for administrative functions	Phase 6
Training Materials	Presentation slides, hands-on exercises	Phase 6

7.2 Software Deliverables

- **Fully Functional USSD Platform:** Complete USSD application with all menu flows, commerce functionality, lay-by system, and extension features
- **Backend API Services:** RESTful APIs for all platform functionality
- **Admin Portal:** React-based web application for system administration
- **Source Code:** Full source code with Git repository access (ownership or perpetual license per contract terms)
- **Database Scripts:** Schema creation, migration scripts, seed data

7.3 Training Deliverables

We will provide comprehensive training for Global Seeds Malawi staff, covering system administration (admin portal usage, user management, inventory management), extension content management (creating and scheduling content, campaign management), payment reconciliation (daily reconciliation

procedures, exception handling), and basic troubleshooting. Training will be conducted on-site at GSM offices with a combination of classroom instruction and hands-on practice sessions.

8. Service Level Agreement

8.1 Support Coverage

Support Level	Coverage Hours	Response Time	Resolution Target
Critical (System Down)	24/7	30 minutes	4 hours
High (Major Feature Impacted)	Business Hours	2 hours	8 hours
Medium (Minor Feature Issue)	Business Hours	4 hours	24 hours
Low (General Inquiry)	Business Hours	8 hours	48 hours

8.2 SLA Inclusions

- Technical support via email, phone, and ticketing system
- Bug fixes and patches for reported defects
- Security updates and vulnerability patches
- 24/7 system monitoring with proactive alerting
- Database backup verification and recovery testing
- Up to 40 hours of minor enhancements/changes
- Monthly system health reports
- Quarterly review meetings

8.3 Uptime Guarantee

InfoVits Ltd guarantees 99.9% system availability, measured monthly, excluding scheduled maintenance windows (communicated at least 48 hours in advance) and force majeure events. Service credits will be provided for any month where availability falls below the guaranteed threshold.

9. Assumptions & Dependencies

9.1 Assumptions

- Global Seeds Malawi will provide timely access to subject matter experts for requirements clarification
- GSM has existing merchant accounts or will initiate the process promptly for Airtel Money and TNM Mpamba
- Product catalog, pricing, and collection point information will be provided during the inception phase
- Extension content (agronomic tips, weather advisory templates) will be provided in both English and Chichewa
- UAT testing will be completed within the allocated timeline with dedicated GSM staff
- USSD short code application will be initiated promptly (typically 4-6 weeks for approval)

9.2 Dependencies

Dependency	Owner	Risk Mitigation
USSD Short Code Approval	GSM/MACRA	Early application; InfoVits to assist with documentation
Mobile Money API Access	Airtel/TNM	Leverage existing InfoVits relationships; parallel setup
Extension Content	GSM	Content templates provided; InfoVits can assist with translation
Collection Point Data	GSM	Required for order fulfillment module; early provision requested
UAT Participants	GSM	Dedicated staff for 2-week testing period

10. Conclusion

InfoVits Ltd is honored to submit this proposal for the Mbeu Zanga USSD Commerce and Farmer Extension Platform. We believe our combination of deep technical expertise, proven track record in USSD and mobile money integration, and intimate knowledge of the Malawian telecommunications landscape make us the ideal partner for this important initiative.

Our proposed solution addresses all requirements outlined in the RFP while incorporating best practices learned from years of developing similar platforms for financial institutions and agricultural programs. The microservices architecture ensures scalability for seasonal peaks, while our comprehensive approach to security protects both farmer data and financial transactions.

We share Global Seeds Malawi's commitment to strengthening food security in Malawi by improving farmer access to quality seed and agricultural knowledge. The Mbeu Zanga platform has the potential to transform how smallholder farmers interact with seed suppliers, making high-quality inputs more accessible and affordable through innovative payment solutions like lay-by.

We look forward to the opportunity to discuss our proposal in detail and to demonstrate how InfoVits Ltd can contribute to the success of this transformative initiative.

Contact Information

Company	InfoVits Ltd
Address	Area 18A , off interchange behind Puma Filling Station, Lilongwe Malawi
Primary Contact	Ngina Kamuloni , Tech Sales manager
Email	info@infovits.com / ngina@infovits.com
Phone	+265 994 89 41 48
Website	www.infovits.com

Thank you for considering InfoVits Ltd as your technology partner.

Appendices

Appendix A: Database Schema Overview

The following provides an overview of the key database entities and their relationships. A complete Entity Relationship Diagram (ERD) will be provided in the Technical Design Document during the inception phase.

Core Entities

Entity	Description	Key Fields
farmers	Registered farmer profiles	id, msisdn, name, district_id, primary_crop, language
products	Seed product catalog	id, name, category, variety, package_size, price, stock
orders	Purchase orders (direct and lay-by)	id, farmer_id, status, total_amount, payment_type
order_items	Line items within orders	id, order_id, product_id, quantity, unit_price
layby_plans	Lay-by payment schedules	id, order_id, deposit_amount, balance, frequency
layby_installments	Individual installment records	id, layby_plan_id, due_date, amount_due, status
payments	All payment transactions	id, order_id, farmer_id, amount, provider, reference
extension_content	Agricultural tips and advice	id, category, title, content_en, content_ny
sms_campaigns	Bulk SMS campaigns	id, name, target_criteria, message_en, message_ny
collection_points	Pickup locations	id, name, district_id, address, contact_person

Appendix B: API Endpoint Summary

The following provides a high-level summary of the REST API endpoints. Complete API documentation in OpenAPI 3.0 format will be delivered as part of the technical documentation.

USSD Gateway Endpoints

Method	Endpoint	Description
POST	/api/v1/ussd/session	Handle USSD session requests from telco gateway
GET	/api/v1/ussd/session/{id}	Retrieve current session state
DELETE	/api/v1/ussd/session/{id}	Terminate session (timeout/exit)

Farmer Management Endpoints

Method	Endpoint	Description
POST	/api/v1/farmers	Register new farmer
GET	/api/v1/farmers/{msisdn}	Get farmer profile by phone number
PUT	/api/v1/farmers/{id}	Update farmer profile
GET	/api/v1/farmers/{id}/orders	Get farmer order history
GET	/api/v1/farmers/{id}/layby	Get farmer lay-by status

Commerce Endpoints

Method	Endpoint	Description
GET	/api/v1/products	List products (with filters)
GET	/api/v1/products/{id}	Get product details
POST	/api/v1/orders	Create new order
GET	/api/v1/orders/{id}	Get order details
PUT	/api/v1/orders/{id}/status	Update order status
POST	/api/v1/layby	Create lay-by plan
POST	/api/v1/layby/{id}/payment	Record lay-by payment

Payment Endpoints

Method	Endpoint	Description
POST	/api/v1/payments/initiate	Initiate mobile money collection
GET	/api/v1/payments/{id}	Get payment status
POST	/api/v1/webhooks/airtel	Airtel Money callback handler
POST	/api/v1/webhooks/tnm	TNM Mpamba callback handler
GET	/api/v1/reconciliation/daily	Daily reconciliation report

Appendix C: Glossary of Terms

Term	Definition
USSD	Unstructured Supplementary Service Data - protocol for mobile communication
Lay-by	Installment payment plan where goods are reserved until full payment
MSISDN	Mobile Station International Subscriber Directory Number (phone number)
API	Application Programming Interface
SLA	Service Level Agreement
UAT	User Acceptance Testing
MACRA	Malawi Communications Regulatory Authority
MERA	Malawi Energy Regulatory Authority
RBM	Reserve Bank of Malawi
TPS	Transactions Per Second

Appendix D: References

- InfoVits Ltd Company Profile and Portfolio - available upon request
- Case study documentation for referenced projects - available under NDA
- Technical certifications and partnership documentation

- Client reference contacts available upon request

— End of Proposal —